

Area in Polar Coordinates

ANSWER KEY



The polar area formula

- 1 State the formula for the area enclosed by a polar curve $r = f(\theta)$ between $\theta = \alpha$ and $\theta = \beta$.

$$A = \frac{1}{2} \int_{\alpha}^{\beta} r^2 d\theta = \frac{1}{2} \int_{\alpha}^{\beta} [f(\theta)]^2 d\theta.$$

- 2 Find the area enclosed by one full loop of $r = 3\cos \theta$ (i.e. $\theta \in [-\frac{\pi}{2}, \frac{\pi}{2}]$).

$$A = \frac{1}{2} \int_{-\pi/2}^{\pi/2} 9\cos^2 \theta d\theta = \frac{9}{2} \int_{-\pi/2}^{\pi/2} \frac{1 + \cos 2\theta}{2} d\theta = \frac{9}{4} \left[\theta + \frac{\sin 2\theta}{2} \right]_{-\pi/2}^{\pi/2} = \frac{9}{4} (\pi) = \frac{9\pi}{4}.$$

Area of a full cardioid

- 3 Find the total area enclosed by the cardioid $r = 1 + \cos \theta$.

$$A = \frac{1}{2} \int_0^{2\pi} (1 + \cos \theta)^2 d\theta = \frac{1}{2} \int_0^{2\pi} (1 + 2\cos \theta + \cos^2 \theta) d\theta. \text{ Using } \cos^2 \theta = \frac{1 + \cos 2\theta}{2}: \\ = \frac{1}{2} \left[\theta + 2\sin \theta + \frac{\theta}{2} + \frac{\sin 2\theta}{4} \right]_0^{2\pi} = \frac{1}{2} (2\pi + \pi) = \frac{3\pi}{2}.$$

Area between two polar curves

- 4 Find the area inside the circle $r = 3$ but outside the circle $r = 2$.

$$A = \frac{1}{2} \int_0^{2\pi} (3^2 - 2^2) d\theta = \frac{1}{2} \int_0^{2\pi} 5 d\theta = \frac{1}{2} (5)(2\pi) = 5\pi.$$

Area of one petal of a rose curve

- 5 Find the area of one petal of $r = 2\sin(2\theta)$, using the interval $\theta \in [0, \frac{\pi}{2}]$ where one petal is traced.

$$A = \frac{1}{2} \int_0^{\pi/2} 4\sin^2(2\theta) d\theta = 2 \int_0^{\pi/2} \frac{1 - \cos 4\theta}{2} d\theta = \left[\theta - \frac{\sin 4\theta}{4} \right]_0^{\pi/2} = \frac{\pi}{2}.$$

Setting up area between curves that intersect

- 6 Set up (but do not evaluate) the integral for the area inside both $r = 2$ and $r = 2 + 2\cos \theta$, noting they intersect where $2 = 2 + 2\cos \theta$, i.e. $\theta = \pm \frac{\pi}{2}$.

By symmetry, for $\theta \in [-\frac{\pi}{2}, \frac{\pi}{2}]$ the circle $r = 2$ is the inner (smaller) boundary, while for $\theta \in [\frac{\pi}{2}, \frac{3\pi}{2}]$ the cardioid is inner. Area = $\frac{1}{2} \int_{-\pi/2}^{\pi/2} 2^2 d\theta + \frac{1}{2} \int_{\pi/2}^{3\pi/2} (2 + 2\cos \theta)^2 d\theta$.

Area of a full rose curve

- 7 Find the total area enclosed by all 4 petals of $r = 3\cos(2\theta)$, using the fact that one petal is traced for $\theta \in [-\frac{\pi}{4}, \frac{\pi}{4}]$.
 Area of one petal: $\frac{1}{2} \int_{-\pi/4}^{\pi/4} 9\cos^2(2\theta) d\theta = \frac{9}{2} \int_{-\pi/4}^{\pi/4} \frac{1+\cos 4\theta}{2} d\theta = \frac{9}{4} [\theta + \frac{\sin 4\theta}{4}]_{-\pi/4}^{\pi/4} = \frac{9}{4} \cdot \frac{\pi}{2} = \frac{9\pi}{8}$. Total for all 4 petals: $4 \times \frac{9\pi}{8} = \frac{9\pi}{2}$.

Area inside a limaçon

- 8 Find the area enclosed by the limaçon $r = 3 + \cos \theta$ (a curve without an inner loop, since the constant exceeds the coefficient).
 $A = \frac{1}{2} \int_0^{2\pi} (3 + \cos \theta)^2 d\theta = \frac{1}{2} \int_0^{2\pi} (9 + 6\cos \theta + \cos^2 \theta) d\theta = \frac{1}{2} [9\theta + 6\sin \theta + \frac{\theta}{2} + \frac{\sin 2\theta}{4}]_0^{2\pi} = \frac{1}{2} (18\pi + \pi) = \frac{19\pi}{2}$.
- 9 Find the area of the small inner loop of the limaçon $r = 1 + 2\cos \theta$ (constant less than coefficient, so an inner loop exists), traced for $\theta \in [\frac{2\pi}{3}, \frac{4\pi}{3}]$.
 $A = \frac{1}{2} \int_{2\pi/3}^{4\pi/3} (1 + 2\cos \theta)^2 d\theta = \frac{1}{2} \int_{2\pi/3}^{4\pi/3} (1 + 4\cos \theta + 4\cos^2 \theta) d\theta = \frac{1}{2} [3\theta + 4\sin \theta + \sin 2\theta]_{2\pi/3}^{4\pi/3}$. At $\frac{4\pi}{3}$: $4\pi - 2\sqrt{3} + \frac{\sqrt{3}}{2}$. At $\frac{2\pi}{3}$: $2\pi + 2\sqrt{3} - \frac{\sqrt{3}}{2}$. Difference: $2\pi - 3\sqrt{3}$. $A = \frac{1}{2} (2\pi - 3\sqrt{3}) = \pi - \frac{3\sqrt{3}}{2} \approx 0.54$.

Setting up area problems (AP-style, no evaluation)

- 10 Set up (but do not evaluate) the integral for the area inside the rose $r = 2\sin(3\theta)$ that lies in the first quadrant only, for one petal traced on $\theta \in [0, \frac{\pi}{3}]$.
 $A = \frac{1}{2} \int_0^{\pi/3} [2\sin(3\theta)]^2 d\theta = \frac{1}{2} \int_0^{\pi/3} 4\sin^2(3\theta) d\theta$.
- 11 Set up (but do not evaluate) the integral for the area between the circle $r = 5$ and the cardioid $r = 5 - 5\sin \theta$ (inside the circle, outside the cardioid), for $\theta \in [0, \pi]$.
 Area = $\frac{1}{2} \int_0^{\pi} [5^2 - (5 - 5\sin \theta)^2] d\theta$.

Finding intersection points of polar curves

- 12 Find all points of intersection of $r = 2\cos \theta$ and $r = 2\sin \theta$ (both circles of radius 1) by setting the equations equal.
 $2\cos \theta = 2\sin \theta \Rightarrow \tan \theta = 1 \Rightarrow \theta = \frac{\pi}{4}$ or $\theta = \frac{5\pi}{4}$. At $\theta = \frac{\pi}{4}$: $r = 2\cos(\frac{\pi}{4}) = \sqrt{2}$. At $\theta = \frac{5\pi}{4}$: $r = 2\cos(\frac{5\pi}{4}) = -\sqrt{2}$ (a negative r , representing the same point reached differently). Both curves also pass through the pole ($r = 0$) at different angles, which must be checked separately.
- 13 Explain why checking intersections of two polar curves algebraically (setting $f(\theta) = g(\theta)$) can MISS the pole as an intersection point, and why it must be checked separately.
 Different curves can reach the pole ($r = 0$) at different values of θ (since a single point, the origin, corresponds to $r = 0$ for ANY θ). Setting $f(\theta) = g(\theta)$ only finds angles where both curves have the SAME r at the SAME θ , so it can miss cases where each curve passes through the origin at a different angle — the pole must always be checked as a separate case.

Area using symmetry to simplify computation

- 14** Explain how the symmetry of $r = 3\cos(2\theta)$ about both axes allows the total area to be found by computing just one petal and multiplying, rather than integrating over the full range at once.

Since $r = 3\cos(2\theta)$ produces 4 congruent petals by symmetry (reflections across both the x - and y -axes map petals onto each other), computing the area of ONE petal via a single definite integral and multiplying by 4 avoids having to integrate piecewise across the full $[0, 2\pi]$ range while tracking sign changes.

Area review: full worked example

- 15** Find the area enclosed by $r = \sqrt{\theta}$ for $\theta \in [0, \pi]$ (a spiral arc closed off by the segment back to the pole).

$$A = \frac{1}{2} \int_0^\pi (\sqrt{\theta})^2 d\theta = \frac{1}{2} \int_0^\pi \theta d\theta = \frac{1}{2} \left[\frac{\theta^2}{2} \right]_0^\pi = \frac{\pi^2}{4}.$$

- 16** Find the area enclosed by $r = 4\sin \theta$ (a circle) for $\theta \in [0, \pi]$, and verify your answer matches the known area formula for a circle of the appropriate radius.

$$A = \frac{1}{2} \int_0^\pi 16\sin^2 \theta d\theta = 8 \int_0^\pi \frac{1 - \cos 2\theta}{2} d\theta = 4 \left[\theta - \frac{\sin 2\theta}{2} \right]_0^\pi = 4\pi. \text{ This matches } \pi r^2 \text{ for a circle of radius 2}$$

(since $r = 4\sin \theta$ traces a circle of radius 2 centered at $(0, 2)$): $\pi(2)^2 = 4\pi$ ✓.

Comparing polar area to rectangular area (conceptual)

- 17** Explain why the polar area formula $A = \frac{1}{2} \int r^2 d\theta$ uses r^2 rather than just r , by relating it to the area of a thin circular sector.

A thin sector of angle $d\theta$ and radius r has area approximately $\frac{1}{2}r^2 d\theta$ (a fraction $\frac{d\theta}{2\pi}$ of the full circle's area πr^2). Summing (integrating) these thin sector areas over the range of θ gives the total polar area formula — the r^2 arises naturally from the geometry of a circular sector, not from any arbitrary convention.

- 18** A student claims the area between $r = 1$ and $r = 2$ (an annulus, full circle) is simply $\frac{1}{2} \int_0^{2\pi} (2 - 1)^2 d\theta$. Explain the error and give the correct setup.

The error is subtracting the radii before squaring — the correct polar formula requires subtracting the SQUARED radii, not squaring the difference of radii. Correct setup:

$$A = \frac{1}{2} \int_0^{2\pi} (2^2 - 1^2) d\theta = \frac{1}{2} \int_0^{2\pi} 3 d\theta = 3\pi \text{ (matching } \pi(2^2 - 1^2) = 3\pi \text{ from the standard annulus-area formula).}$$